

BIRLA INSTITUTE OF TECHNOLOGY MESRA
OFF CAMPUS DEOGHAR
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Machine Learning-Based Semiconductor Manufacturing Process
Pass/Fail Prediction Using Random Forest,
Gaussian Naive Bayes, and Support Vector Machine

A PROJECT REPORT

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Project Guide

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CERTIFICATE

**This page is reserved for the official certificate issued by the Department of Computer Science and Engineering, Birla Institute of Technology Mesra, Off Campus Deoghar.
The official certificate will be inserted before the final submission of this project report.**

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ABSTRACT

The semiconductor manufacturing industry generates a large amount of sensor data during the production process. Analysing this data helps manufacturers monitor product quality and identify whether a manufactured semiconductor device meets the required quality standards. Predicting the pass or fail status of manufactured products at an early stage can significantly reduce production costs, minimize defects, and improve overall manufacturing efficiency.

The objective of this project is to develop a Machine Learning model capable of predicting the pass or fail status of semiconductor manufacturing processes using sensor data. The dataset used in this project consists of numerous process variables collected from different stages of semiconductor manufacturing. Since the dataset contains missing values and an imbalanced target class, appropriate data preprocessing techniques were applied before model development.

The preprocessing phase included handling missing values, removing highly incomplete features, balancing the dataset using the Synthetic Minority Oversampling Technique (SMOTE), and standardizing the feature values using StandardScaler. After preprocessing, three supervised Machine Learning algorithms—Random Forest, Gaussian Naive Bayes, and Support Vector Machine (SVM)—were implemented and evaluated.

To improve model reliability, five-fold Cross Validation was performed for each model. Furthermore, GridSearchCV was used to optimize the hyperparameters of the Support Vector Machine. The optimized SVM model achieved the highest prediction accuracy among all implemented models, demonstrating its effectiveness for semiconductor quality prediction.

This project illustrates the importance of proper data preprocessing, feature preparation, model evaluation, and hyperparameter tuning in developing an efficient predictive system for semiconductor manufacturing quality assessment. The trained model was finally saved for future prediction and deployment purposes.

GitHub Repository:

<https://github.com/parammeet83/Semiconductor-Manufacturing-Pass-Fail-Prediction>

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1. INTRODUCTION

1.1 Introduction

Semiconductor manufacturing is one of the most advanced industrial processes, involving hundreds of sensors that continuously monitor various production parameters. These sensor readings help manufacturers evaluate whether a semiconductor wafer satisfies the required quality standards. Even a small variation in these parameters may result in a defective product, leading to increased manufacturing costs and reduced production efficiency.

Machine Learning provides an effective solution for analyzing large volumes of manufacturing data and predicting product quality. Instead of manually examining numerous sensor readings, Machine Learning algorithms can automatically identify patterns associated with successful and defective manufacturing processes.

In this project, a Machine Learning-based prediction system was developed using the Semiconductor Manufacturing Process dataset. The dataset was pre-processed by handling missing values, removing highly incomplete features, balancing the target classes using SMOTE, and scaling the features using StandardScaler. Three supervised Machine Learning algorithms—Random Forest, Gaussian Naive Bayes, and Support Vector Machine (SVM)—were implemented and evaluated. Cross-validation and hyperparameter tuning using GridSearchCV were performed to improve model performance. The optimized Support Vector Machine achieved the highest prediction accuracy and was selected as the final model.

1.2 Problem Statement

The semiconductor manufacturing process generates a large number of sensor measurements for every manufactured product. Due to the complexity of the production process, identifying defective products manually is time-consuming and may lead to errors. The objective of this project is to develop a Machine Learning model capable of accurately predicting whether a semiconductor manufacturing process will result in a pass or fail outcome using sensor data.

1.3 Objectives

- To analyze the semiconductor manufacturing dataset.
- To preprocess the dataset by handling missing values and removing highly incomplete features.
- To balance the dataset using SMOTE.
- To standardize feature values using StandardScaler.
- To build prediction models using Random Forest, Gaussian Naive Bayes, and Support Vector Machine.
- To evaluate model performance using accuracy and cross-validation.
- To optimize the Support Vector Machine using GridSearchCV.
- To identify the best-performing model for semiconductor pass/fail prediction.

2. METHODOLOGY

2.1 Dataset Description

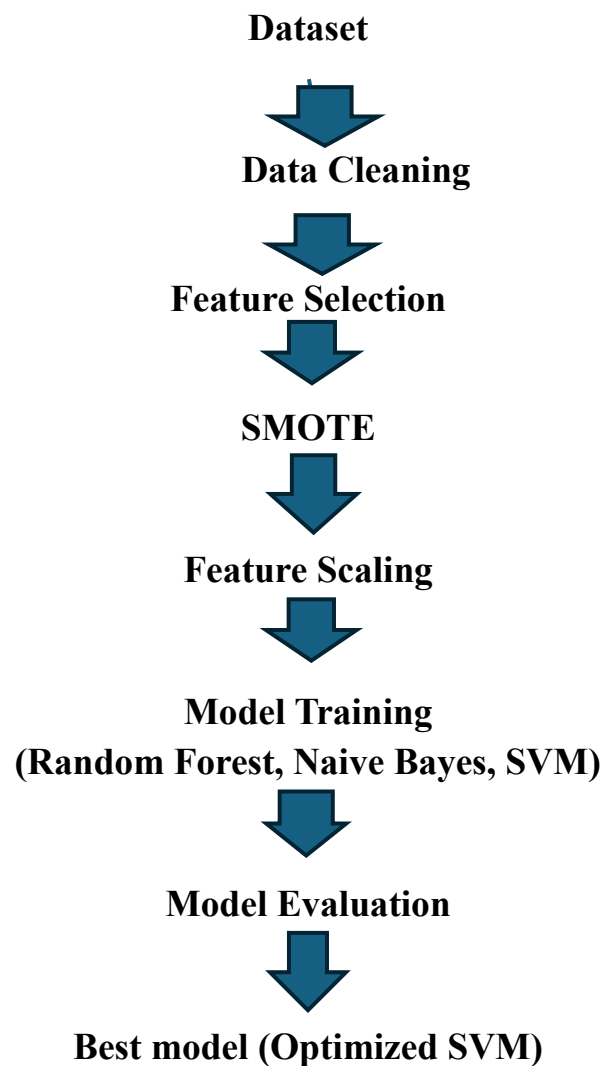
The dataset used in this project contains sensor measurements collected during the semiconductor manufacturing process. It consists of 1,567 observations and 592 attributes, including one timestamp column, 590 sensor features, and one target column (Pass/Fail). The target variable indicates whether a manufactured semiconductor product passed quality inspection or failed. Before model development, the dataset was analysed to identify missing values, data types, and class imbalance.

2.2 Data Preprocessing

The following preprocessing steps were performed:

- Removed the Time column as it was not useful for prediction.
- Removed features containing more than 55% missing values.
- Filled the remaining missing values using the median.
- Balanced the dataset using SMOTE.
- Standardized the feature values using StandardScaler.
- Split the dataset into 80% training and 20% testing sets.

2.3 Workflow



3. MODEL DEVELOPMENT

Three supervised Machine Learning algorithms were implemented to predict the pass/fail status of semiconductor manufacturing products. The dataset was divided into training and testing sets using an 80:20 ratio. Model performance was evaluated using classification accuracy and five-fold cross-validation. Hyperparameter tuning was performed using GridSearchCV to improve the performance of the Support Vector Machine.

3.1 Machine Learning Models Used

Model	Purpose
Random Forest	Baseline ensemble classifier for handling high-dimensional data
Gaussian Naive Bayes	Probabilistic classifier used for comparison
Support Vector Machine (SVM)	Final optimized model with the highest performance

3.2 Model Evaluation

The trained models were evaluated using testing accuracy and five-fold cross-validation. GridSearchCV was applied to optimize the hyperparameters of the Support Vector Machine. The optimized SVM achieved the highest testing accuracy and was selected as the final prediction model.

4. RESULTS

Model	Test Accuracy	Cross Validation Accuracy
Random Forest	92.99%	99.36%
Gaussian Naive Bayes	25.48%	60.30%
Support Vector Machine	93.31%	99.57%
Optimized SVM	93.63%	99.62%

The optimized Support Vector Machine achieved the best overall performance with a testing accuracy of **93.63%** after hyperparameter tuning using GridSearchCV. Random Forest also performed well, while Gaussian Naive Bayes showed comparatively lower performance due to the characteristics of the dataset.

5. Conclusion

This project successfully developed a Machine Learning model for predicting semiconductor manufacturing pass/fail outcomes using sensor data. Data preprocessing, SMOTE, feature scaling, and model optimization significantly improved prediction performance. Among the evaluated algorithms, the optimized Support Vector Machine provided the highest accuracy and was selected as the final model.

6. Future Scope

- Explore Deep Learning models for improved prediction.
- Apply advanced feature selection techniques.
- Deploy the model as a web application.
- Evaluate the model on larger industrial datasets.

PROJECT REPOSITORY

The complete implementation of this project, including the source code, Jupyter Notebook, trained machine learning model, project report, presentation, and other supporting files, is available in the GitHub repository below.

GitHub Repository

<https://github.com/parammeet83/Semiconductor-Manufacturing-Pass-Fail-Prediction>

The GitHub repository contains the complete implementation, trained model, documentation, report, and presentation for this project.